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Rain as Commons:

Rainwater Harvesting and Collective Self-Sufficiency in Rural Catalonia

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Built Environment
A bachelor's degree thesis, 18 hp / 18 ECTS
Spring 2026
Supervisor: Ana Betancour
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RAIN AS COMMONS:

RAINWATER HARVESTING AND COLLECTIVE SELF-SUFFICIENCY IN RURAL CATALONIA

An Architectural and Participatory Study of Off-Grid Water Management in a Queer Affinity Collective

Thesis project by:

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ABSTRACT

This thesis investigates how rainwater harvesting (RWH) can function not only as a technical system, but as a socio-material practice supporting self-sufficiency and resilience in an off grid, queer affinity collective in rural Catalonia, Spain. Framed through the concept of water as a commons by economist Elinor Ostrom (1990[2011]), the study examines how decentralised water infrastructure can be locally adapted and collectively governed within a non-hierarchical community.

The research employs a mixed-methods approach combining literature review, fieldwork, participatory design, and water balance modelling within a Participatory Action Research (PAR) framework. Initially, the project proposed a low-tech Biosand filtration system (BSFS) for potable water treatment. The participatory process revealed that, despite its technical feasibility, the system was perceived as too ´scary´ and unbalanced, as responsibility would fall on only a few people. Concerns about uneven distribution of responsibility significantly affected its acceptance within the community. Rather than implementing the original design, the project pivoted toward a participatory intervention that transformed existing water tanks. With collective workshops that enabled the community to appropriate the infrastructure as a shared cultural artifact. This process strengthened engagement, ownership, and everyday interaction with the system.

The findings demonstrate that rainwater can be governed as a commons when infrastructure is visible, collectively managed, and culturally integrated. Governance in this context relies on mutual care, shared responsibility, and non-hierarchical forms of accountability rather than formal enforcement mechanisms.

The thesis concludes that RWH in off-grid collectives must be understood as both a technical and social process. It contributes by reframing RWH in the Global North as a socio-material commons, demonstrating how Ostrom’s design principles can be adapted to non-hierarchical settings through passive monitoring, restorative accountability, and participatory appropriation of infrastructure.

KEYWORDS

Rainwater Harvesting, Water commons, Self-sufficiency, Off grid, Participatory design

ACKNOWLEDGEMENTS

I would like to thank all dear friends and family who contributed towards this project through their generous support of time, tools, and funding. Especially to Loke Läck and Jesús Duriez, for their constant interest, encouragement, and support throughout the process.

A thank to my supervisor, Professor Ana Betancour, for guiding my ambitious ideas through moments of chaos.

Finally, my deepest thanks to the Affinity Group. You show that family can be chosen and continually remind me that our strength lies in togetherness, and that no resource is more vital than love, understanding, and collective care.

A WORD FROM THE RESEARCHER

My journey into this research began not in a library, but through chance I encounter the Catalan affinity collective during my recent time in Spain for work. Shared friends within the same social milieu opened the door to their rural settlement, where I first witnessed the daily struggles of living off grid. But also, the appeal of living self-sufficiently within a close community. What started as casual conversations evolved into collaborative fieldwork, revealing water scarcity a main reason, holding back their self-sufficient aspirations. This personal entry shaped the project's dual focus: Rainwater harvesting and Water as a commons.

As a third-year architecture student drawn to decentralised systems, I approach this case with enthusiasm for low-tech solutions and alternative lifestyles. Also, reflexivity about my position, with risks of romanticising off-grid living. While other problems which arise for communities over time, cannot be experienced in my situation as a visitor.

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DEFINITIONS

Rainwater Harvesting (RWH) - Sometimes called “runoff harvesting”, is the catchment of rainwater runoff, either through reshaping of landscape (macro- catchment), or from example rooftops or courtyards (micro- catchment), collecting water via gutters to cisterns. (Beckers et al. 2013)

Affinity Group - A small collective united by shared affinities: such as values, lifestyles or skills. Often working along the principles of non-hierarchy, autonomy, participation and direct democracy (Seeds For Change n.d). Sociologist Max Weber drew on the author’s Goethe's concept of ´Elective Affinities` from 1807 to describe non-deterministic attractions between ideas, values, and social structures that bind groups together. (McKinnon 2010) Thus, being a well-suited synonym for the research’s case group, besides the more used definition ´collective´.

Common-Pool Resource (CPR) & Commons - CPR is a subtractable natural resource, that is rivalrous in use and hard to exclude others from, making it prone to overexploitation unless collectively governed. Some examples of CPRs are fishing grounds, groundwater basins, grazing areas, bridges, streams and oceans. Ostrom (1990[2011]) presents that Commons (shared resources) can remain prosperous through communal decision-making, participation and monitoring. Explained further in Chapter 3.2 ´Water as a Commons´.

Off grid - Living independently from public utilities, generating on-site power, water and waste-systems. Off grid describes the context of the collective.

Resilience (Climate Resilience) - The ability to prepare for, recover from, and adapt to the impacts of Climate Change. `Resilience` is first referenced in the 17th century by Francis Bacon as the characteristics of an echo. The ability of bouncing back from something. To be Climate Resilient, Risk must be evaluated. Risk = Vulnerability x Threat (C2ES 2019, p. 1-3).

Self-sufficiency - The state of meeting one’s own essential needs such as food, water and energy. Emphasising independency and autonomy. In the case of the research, self-sufficiency also regards to being secure in the rural setting in which the collective lives.



Figure 1: Arrival on site. Analogue photograph by author.

1 INTRODUCTION

Water scarcity and off-grid living create challenges for communities that depend on decentralised systems for everyday survival. In the Catalan affinity collective, these challenges are closely tied to the ways water is accessed, shared, maintained, and governed within a non-hierarchical social structure. This makes rainwater harvesting (RWH) not only a technical question, but also one of organisation, responsibility, and collective care. Rather than treating RWH as a one-time architectural intervention, this project examines how rainwater can be continuously resourced, managed, and governed as a shared commons.

This investigation emerges in response to a growing global water crisis. A recent report has described the current situation as one of "global water bankruptcy" (UNU-INWEH, 2026), driven by the intersecting pressures of climate change, population growth, political instability, and rising living costs. These conditions are increasingly pushing people toward off-grid and informal ways of living, particularly in peri-urban and rural areas where access to reliable water infrastructure is often limited or failing. In such contexts, water scarcity is not only a technical issue, but a social and spatial challenge that affects everyday life. In the Mediterranean countryside, characterised by irregular rainfall patterns and prolonged droughts, decentralised water systems offer critical strategies for resilience. RWH presents one such approach. When harnessed through appropriate design, rainwater becomes a freely available common resource with potential to enhance self-sufficiency.

This research focuses on a specific case: an off-grid collective of 50 members in rural Catalonia, whose current situation revolves around fetching and transporting water from nearby rivers and towns. The dependency on distant water sources raises vulnerability and social stress for the community. The collective's dynamic and non-hierarchical social structure creates an opportunity to explore rainwater harvesting systems (RWHS) as collectively managed resources through Elinor Ostrom's (1990[2011]) theory of Common-Pool Resource governance. As the participatory research process unfolded, the focus of the thesis shifted toward analysing how such systems can be meaningfully introduced and locally adapted to the specific conditions of the site, while also examining how water can be governed as a shared commons in practice.

The research is divided into nine parts: (1) Introduction with Problem Background, Aim and Research Questions (2) Method, (3) Theory / Technical and Theoretical Framework, (4) Case and Context, (5) Initial Design Proposal, (6) Results, (7) Discussion and Conclusion, (8) References & (9) Appendix with technical calculations and documented workshops.

1.1 PROBLEM BACKGROUND

1.1.1 Global Water Scarcity

Water covers about 71% of the Earth's surface, yet freshwater suitable for human use accounts for less than 1% of the total. Water is essential for all known living organisms and is the most basic need for human health, sanitation, and well-being. Today, however, the world has entered the stage of *Global Water Bankruptcy*, where decades of overexploitation have resulted in a state of post-crisis, where water reserves have been eroded to the point, where simple restoration is no longer possible (UNU-INWEH 2026, p. 10, 15, 43). There is 2.2 billion people who lack managed drinking water and 4 billion face severe water scarcity at least one month per year (UNU-INWEH 2026, p. 13).

Due to climate change, increasing consumption, industrial growth and rapid land use changes, water demand is expected to rise exponentially. Affecting water supply, intensifying water shortages and worsening water quality. Especially in rural sites and for vulnerable communities (Joji & Jacob 2023, p. 5; UNEP 2009, p. 1).

Using climate models and economic scenarios, the World Resources Institute ranked water stress in 167 nations. They found 33 countries will face extremely high water-stress by 2040. Spain being amongst them, estimated to extract more than 80% of surface water annually (Figure 2) (Gassert et al. 2013). More recent reports from UNU-INWEH (2026) shows that water scarcity is more a global issue than regional (Figure 3), with high overall risk across the globe. Estimated changes of droughts however, present Spain among the most vulnerable regions, estimated to increase 350% from previous decades. (Figure 4)

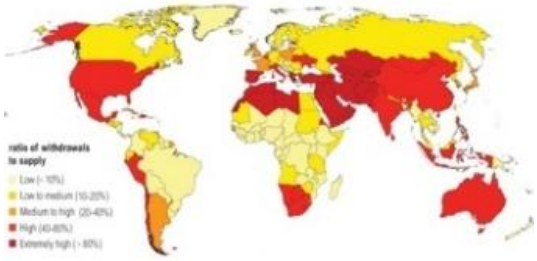


Figure 2: Water Stress by Country 2040 (Gessert et al. 2013).

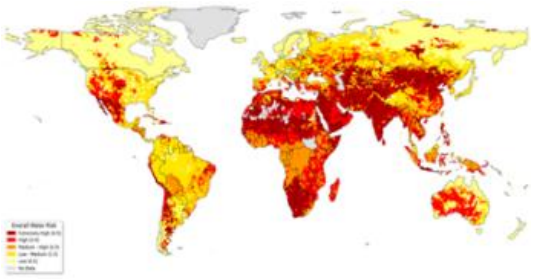


Figure 3: Overall Water Risk across the Globe (UNU-INWEH 2026, p. 33).

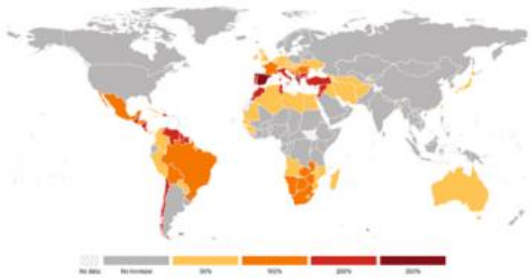


Figure 4: Expected Changes in Drought Occurrence Likelihood. (UNU-INWEH 2026, p. 31).

1.1.2 Rural Living and Informal Settlements

"*One billion people live in informal settlements in the world. A number projected to triple by 2050"* (Sachdeva et al. 2025). Informal settlements are marked by hazardous environments, overcrowded living conditions and limited access to services such as clean water. Although communities need resilience-strategies, they are hard to introduce from a community level due to low resources. But also, from a top-down perspective, which often fail to address the needs accurately. Community-led climate adaption strategies prove to have better impact than top-down initiatives. (Sachdeva et al. 2025)

Alongside urbanisation, there is a trend of people moving or being displaced into peri-urban and rural areas. Often driven by economical-, legal-, or climate-related reasons. Rural living also attracts people searching for an alternative off-grid living situation. This includes homesteads, off-grid communities, squatters, travelling communities and residents of temporary shelters, eco-villages and ´Earthships´.

1.1.3 Growing Role of Rainwater Harvesting

Current global water scarcity leads to the growing need of non-conventional water systems. To reduce stress on existing systems and combat climate-driven water shortages (Khanal et al. 2020). These developments make adaptive water management essential under new environmental and socio-economic conditions. RWH plays a key role by improving water security. Turning rainwater runoff into a local resource. (El-Sayed et al. 2010).

Although its long history, RWH has just recently been identified as a comprehensive source of alternative water supply. Some reasons to why it's not more common is due to individuals being uninformed about the *know-how* and having doubts about provided water quality. (Damman et al. 2017). On a larger scale it is also because centralised water providers who operate larger regions, ignore RWH as a possible system (Che-Ani et al. 2009, p. 134) as many municipalities depend entirely on groundwater (Joji & Jacob 2023, p. 2).

The growing interest in RWH is reflected in newer water policies all over the world. In recent decades RWH has become cheaper and more predictable in performance. (Thomas & Martinson 2007, p. 9)

1.2 AIM AND RESEARCH QUESTIONS

This thesis investigates how RWH can support self-sufficiency and resilience in an off-grid affinity collective in rural Catalonia and analysing how RWH functions as an active socio-material practice. The research examines how decentralised water systems can be technically adapted and socially integrated within a non-hierarchical community, and how rainwater can be governed as a shared commons through collective responsibility, participation, and care.

Further, the research examines how Ostrom's (1990[2011]) design principles can be adapted to a non-hierarchical collective, with particular attention to how monitoring and sanctions may require alternative approaches within the affinity group. Addressing this topic reveals a significant research gap, particularly in the *´Global North´*, where RWH is frequently framed as a technical, cost-saving add-on rather than a foundational method of livelihood. Consequently, there is a pressing need to understand how decentralised water systems are tangibly integrated into collective everyday routines and physical space.

To bridge this gap, the research pursues a multidisciplinary approach, combining architecture, environmental science and participatory design, to explore RWH as a viable, locally controllable resource in drought-prone rural environments. The research questions were refined as the participatory process revealed the social implications of introducing potable water infrastructure.

1. *How can rainwater harvesting strategies be introduced and adapted for self-sufficiency in an off-grid Catalan collective, considering both technical feasibility and social acceptance?*

2. *How can rainwater be collectively governed as a commons within the affinity group's social structure to enhance shared responsibility and resilience?*

1.3 EXISTING RESEARCH

Existing research on RWH and off-grid living has grown significantly in recent years. Research regarding RWH has also broadened, covering new aspects, such as urban agriculture, economic analysis, water-energy nexus and urban flood-mitigation using advanced computer modelling tools. (Rahman 2018, p. 2-4).

A systematic review of over 80 studies on small-scale rainwater harvesting systems (RWHS) by Nandi & Gonela (2022) highlights that research focus on technical design parameters, such as storage-tank sizing, demands-supply-modelling and impact on climate change. The systematic research present three key findings, which all will be to great importance for this research: (Nandi & Gonela 2022, p. 1, 8)

(1) Although RWH is intended to create social benefits, existing literature does not sufficiently support social policymaking or management of rainwater as a public utility.

(2) Policies, need to take socio-economic and socio-environmental realities into account to achieve successful RWH systems.

(3) Modelling RWH systems is complex, considering spatial differences, local policy evaluations and theories of governance.

Existing RWH- literature also reveals a distinct geographical difference. In the *´Global North´*, RWH is framed as an optional climate adaptation. Such as stormwater management, analysed through yield, return on investment, and energy efficiency. This technical focus largely overlooks small-scale, communal water governance. Conversely, *´Global South´*- research is often driven by a humanitarian water- scarcity frameworks. Focusing on low-tech interventions based on health and cost-efficiency. Often prioritising large-scale agricultural infrastructure in arid regions.

Few studies examine how these systems are socially negotiated in informal, non-hierarchical settings, and academic literature often overlooks decentralised and informal living communities. As a result, communities rely on DIY tutorials and video-sharing platforms for the most practical, adaptable, and low-tech solutions.

2 METHOD

2.1 METHODOLOGY OVERVIEW

The methodological approach is investigative, experimental, and participatory. It combines qualitative methods such as fieldwork, observation, participatory workshops, and informal conversations, with quantitative elements including water balance modelling and climate data. This mixed approach was chosen because the research addresses both the technical performance of RWH and its deep social significance. Since the collective is non-hierarchical and operates with limited resources, the methodology was required to remain flexible, low-cost, and responsive to the realities of the site.

Initially, this research aimed to design and implement a low-tech Biosand filtration system for the collective. Detailed drawings, approximate budgets, construction timeframes, and maintenance schedules were developed as part of the technical proposal. However, fieldwork revealed that the collective had lost interest in managing a potable filtration system. Instead, they preferred to expand their current, simpler method of direct roof-to-tank collection for non-potable use.

2.2 PARTICIPATORY ACTION RESEARCH

Participatory Action Research (PAR) provided an appropriate guiding framework. PAR focuses on responding to local preferences and producing knowledge through collaboration. (University of Reading 2023). This approach is especially important in a community context, as local collaboration and daily routines directly determine whether any water system can function in practice.

Changing focus of the water tanks from technical infrastructure into a participatory art project. The research changed focus based on the community's lived reality. Instead, exploring how infrastructure can be socially integrated, visually reclaimed, and transformed into a shared symbol of the water commons, ultimately raising local interest for future development.

To complement PAR, the research also took inspiration from Participatory Rural Appraisal (PRA). While PAR is strongly centred on collective action and social change through reoccurring cycles of reflection and intervention, PRA contributes a more practice-based set of methods for revealing local knowledge. It aims to empower locals to analyse their own living conditions and plan actions through accessible methods like mapping, communal walks, and group discussions (Sontakki, Venkatesan & Rao 2019).

The methodology is structured into three phases: Phase 1: Literature Review, Phase 2: Fieldwork & Documentation & Phase 3: Participatory Design Process & Construction.

2.3 PHASES OF RESEARCH

2.3.1 Phase 1: Literature Review

The literature review establishes the theoretical and technical foundation for the thesis. It brings together RWHS, vernacular and historical water practices, commons governance, and off-grid living and social acceptance to name a few. The purpose of this phase was to identify both technical strategies and conceptual tools that could inform the design research.

Key technical sources include *Designing Rainwater Harvesting Systems*, (2014) by Novak, Van Giesel & Debusk, which provides a clear framework for integrating rainwater into building systems. *Design for Water: Rainwater Harvesting, Stormwater Catchment and Alternate Water Reuse*, (2007), by Kinkade-Levario, which offers both contemporary and historical examples of rainwater systems at different scales. Terry Thomas and Brett Martinson's *Roofwater Harvesting: A Handbook for Practitioners,* (2007), was also particularly useful because of its practical, construction-oriented guidance and its strong focus on implementation. Together, these sources helped establish the technical logic of RWH, including components, sizing, storage, filtration, first flush systems, and maintenance.

In parallel, Elinor Ostrom's *Governing the Commons*, (1990[2011]) is a central theoretical source for the governance dimension of the thesis. It provides the basis for understanding rainwater not just as a material input, but as a shared resource that requires collective rules and governance.

Another important inspiration for the research is from the book *Research for People Who (Think They) Would Rather Create* by Dirk Vis (2025), who presents how multiple artistic expressions can be combined in research. Which during the research pivot, inspired to broaden the view of the research and to question the actual purpose of the project.

The literature review of books and articles also covers broader ideas of vernacular practice, climate adaptation, low-tech design and self-sufficient living.

2.3.2 Phase 2: Fieldwork & Documentation

Fieldwork and documentation were used to understand the collective's existing water practices, spatial organisation and the site. This phase combined ethnographic observation, mapping, photography, sketching, and informal conversations. The aim was to understand how water is currently used, where it comes from, how it is moved, and how it fits into daily life and collective routines.

The fieldwork took place during two site visits over three weeks in total: one week in January and two weeks in April. These visits allowed for a broader understanding of the site, its weather conditions, and the collective's changing needs.

2.3.3 Phase 3: Participatory Design Process & Construction

The participatory design process took place through weekly communal meetings, workshops, and informal conversations during work hours. Participation remained open and informal throughout the process, allowing people to join when they were available and contribute according to their interest and capacity.

The first aim of this phase was to discuss needs, present possible design options, and explore what kind of RWHS made sense for the collective. This included introducing the initial proposal of the Biosand filter. However, as the process unfolded, it became clear that the collective did not want to take on a potable filtration system (at that moment). This prompted a methodological pivot in which the project was adapted to the community's actual preferences and practices.

The objective of this phase was to embed Ostrom's principles of collective governance into the technical solution while ensuring that the design fit the affinity group's social structure.

The final days of site-visit was for testing and evaluating the adapted intervention.

2.4 ETHICAL CONSIDERATIONS

Research with the Catalan collective balance's academic rigor with respect for their autonomy and privacy. Attention was also given to power dynamics throughout the research, particularly in relation to access, representation, and the researcher's position as an outsider. Because their internal gatekeeping dictates the consent process, strict confidentiality is applied.

Ethical considerations include anonymising all identifiable markers such as names, license plates, tattoos, and precise geographic locations. Although the collective maintains an open public profile, these practices align with Swedish research-ethics guidelines on *god forskningsed* (Swedish Research Council 2024, p. 24–27, 63–70), as well as conducting and presenting an honest and transparent research across all phases of the study (pp. 84–85).

Finally, regional climate and legislative data are intentionally generalised to safeguard the site while ensuring the study remains broadly applicable.

3 THEORY

3.1 RAINWATER HARVESTING

Water harvesting is the combined practice of collecting and storing water from various sources, including groundwater, surface water, and atmospheric water. Rainwater harvesting (RWH) collects runoff from rain-fallen surfaces for domestic, agricultural, or animal use. These systems are typically categorised by their catchment type: micro-catchment (e.g., rooftops and courtyards) and macro-catchment (e.g., hillside conduits) (Figure 5) (Beckers, Berking, & Schütt 2013. p. 147).

RWH is a decentralised water system, which allows users to manage both demand and supply directly. It is further easy to construct, maintain, operate and expand over time (Kinkade-Levario 2007, p. 13). Depending on household needs, RWH can be implemented as a single, main, or solely potable water source. Relying on RWH as a single source provides great self-sufficiency but requires year-round rainfall, large storage capacity, and substantial roof runoff area, which can be highly expensive. Conversely, using RWH as a main source allows for smaller tanks, though it carries the risk of overflow during rainy seasons and running dry during periods of droughts. To mitigate this, users can employ adaptive demand strategies, periodically lowering water consumption based on current storage levels (Thomas & Martinson 2007, p. 28).

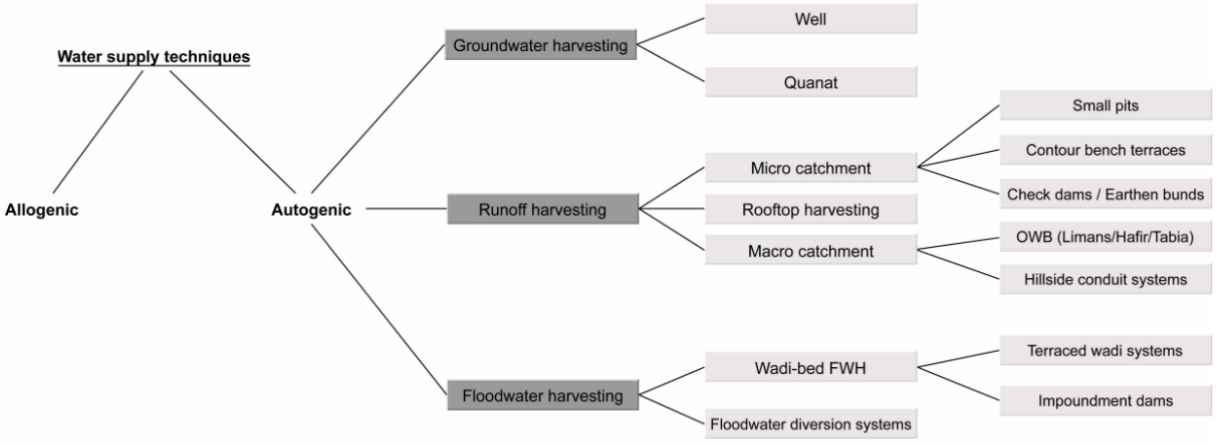


Figure 5: Classifications of Water Harvesting Methods. (Beckers, Berking & Schütt 2013, p. 147).

3.1.1 Vernacular & Historic Context



Figure 6: Examples of Vernacular Rainwater Harvesting techniques. Figures A-L listed separately in list of figures. Collage by author.

Historical evidence demonstrates that finite water resources have always required careful management and shaped both the rise and decline of civilisations (Novak 2014, p. 25). Rainwater harvesting systems (RWHS) are not a recent innovation, but a historically embedded practice, with documented origins in Mesopotamia around 5,000 years ago (Agarwal & Narain 2000; Prinz 2000). Across arid and semi-arid regions, these systems were essential for sustaining long-term habitation. For example, 3,000 years ago in Yemen, runoff harvesting irrigated 20,000 acres, feeding 300,000 people (Joji & Jacob 2023, p. 4–5). Traditional systems in Pakistan such as *Karez* and *Sailaba* are still in use. As are pre-roman systems from northern Africa, such as the *meskat, jessour* and *wadi-beds* (Kinkade-Levario 2007, p. 7; Joji & Jacob 2023, p. 4-5). Similarly, in the Minoan civilization (3200–1100 BC), cities such as Phaistos collected rainwater from rooftops and courtyards, and these systems included first-flush diversion and sedimentation to improve water quality (Angelakis et al. 2026, p. 3; Angelakis & Spyridakis 1996, pp. 163–174).

In a Spanish context, evidence of RWH in vernacular architecture is comparatively limited, with only 9 out of 212 documented historical buildings integrating such systems, primarily located in the Balearics (Cañas & Martín 2004). However, these cases demonstrate a high degree of climatic adaptation, suggesting that even infrequent applications were carefully tailored to local environmental conditions.

These examples show that local knowledge allowed communities to adapt water systems to local environmental conditions over long periods, often with a high level of technical sophistication. Taken together, they position RWH as a historically grounded practice of climate adaptation and collective resource management, helping ensure water access and supporting prosperity and growth.

3.1.2 Community-based Rainwater Harvesting Systems as a Sustainable Practice

While the modern market prioritises centralised infrastructure, it is both economically and ecologically more sustainable to process rainwater locally (Partzsch 2009, p. 985–986; Kinkade-Levario 2007, p. 13). Community-based rainwater harvesting systems (CBRWHS) differ from individual household systems because they are collectively planned, built, and managed in shared spaces, emphasising collective ownership and shared responsibility (Mukarram 2023, p. 2). They are particularly effective for domestic supply in rural communities, where ways of living are deeply tied to the local landscape (UNEP 2009, p. 63). RWH provides not only a local source of high-quality water, but also broader environmental benefits and greater resilience to drought and other hazards (UNEP 2009, p. 63-65; Che-Ani et al. 2009, p. 137). Furthermore, studies indicate that implementing RWHS yields positive social outcomes, such as improved health, gender equality, and strengthened group coherence (Damman et al. 2017, p. 42–47; UNEP 2009, p. 65; Agarwal & Narain 2000). Ultimately, Partzsch (2009, p. 985–987) argues that decentralised RWH systems remain underdeveloped and rare in practice and calls for a reversal of traditional paradigms that gives new priority to climate resilience and local adaptability.

A decentralised water system, however, has special management requirements. Previous cases evaluated by Agarwal & Narain (2000) show that CBRWHS perform better than large-scale water development projects, which have often led to inefficient and unequal water distribution and even forced resettlements. To achieve community-led water management, participatory democracy must be strengthened and people's participation at village or community level must grow. They highlight several steps towards this: maintaining water as a community resource, adopting an integrated approach to local resource development, and promoting decision-making forums (Agarwal & Narain 2000).

3.1.3 Water as a Tool of Cooperation

Throughout history, water has functioned both as a source of conflict and as a catalyst for cooperation (Prinz 2000). Although water scarcity is a global issue, water-related risks have historically led more to cooperation than conflict (UN Water 2022) and have long played a unifying role in society. Increasing evidence shows how CBRWHS and watershed management strengthen social capital, having a positive impact on developing further decentralised systems (UNEP 2009, p. 9). In such contexts, water becomes not only a material necessity, but a structuring force shaping social relationships, institutions, and collective practices (Vinciguerra 2024, p. 954). For off-grid collectives, this dynamic is especially relevant. Without access to centralised infrastructure, cooperation becomes a prerequisite for survival. For the Catalan collective, water has further potential to become a social medium through which cooperation is organised and maintained in daily life.

Importance for research

This theoretical framework positions RWH as both a technical system and a socially embedded practice, showing that its effectiveness depends on more than engineering performance. It highlights how water has long functioned, and can still function, as a tool for collaboration, and it provides the basis for examining RWH as a historically grounded, community-based resource for self-sufficiency and resilience within a non-hierarchical collective. At the same time, it narrows the study to atmospheric water and micro-catchment rooftop systems, while also distinguishing between RWH as a single source, main source, or potable-only source. Taken together, these perspectives are essential for analysing the Catalan affinity collective, where technicality and social structures go hand in hand.

3.2 WATER AS A COMMONS

In contemporary environmental design and theory, there are numerous paradigms through which we can understand and relate to water. Institutional frameworks often treat water as an engineered utility or an economic commodity, while legal rights organisations approach water as a fundamental human right. In the realm of sustainable design, methodologies such as Circularity and Cradle to Cradle, focus on closing resource loops and minimising ecological impact. Broader sociopolitical concepts, such as ´Water Democracies´, further highlight the ties between water, power, and society. While these diverse approaches provide crucial insights, this research centres on Water as a commons. This perspective emphasises collective governance, providing the theoretical foundation for how communities can autonomously manage their local water resources.

Significant academic differences exist regarding definitions such as ´Public good´, ´Common good´, ´Collective good´ & ´Commons´. Depending especially on the user, if it be from an ´economic background´, they tend to relate to financial sustainability, investment and alternative financing or, from other backgrounds such as lawyers, human rights activists or representatives of states who tend to relate to water being a vital necessity. (Vinciguerra 2024, p. 949-954). This research will use the definition commons, based on common-pool resource (CPR) by Nobel Prize winner in economics, Elinor Ostrom, who in 1990, *Governing the commons: The Evolution of Institutions for Collective Action*. Argues that water-systems, such as irrigation systems and groundwater aquifers, works best when collectively managed, rather than purely state- or market controlled.

A central argument by Ostrom is that CPRs do not inevitably fall into the “tragedy of the commons” classically described by ecologist Garrett Hardin. Instead, many communities develop their own rules, monitoring practices, and sanctions that allow them to use shared resources sustainably over time. (Ostrom 1990[2011], p. 180-183)

Although Ostrom did not explicitly categorise RWH as a CPR, rainfall was consistently addressed as an uncertain natural phenomenon affecting water access in her case studies. Particularly the Spanish *huertas* (Ostrom 1990[2011], p. 69-82). *Rainwater as a commons* transforms an unutilised resource into a managed supply. However, it requires the collective to integrate it into their spatial and social organisation through adapted design principles.

3.3 DESIGN PRINCIPLES FOR GOVERNING COMMON-POOL RESOURCES

Through detailed case studies, Ostrom demonstrates how local actors successfully co-manage CPRs by defining common rules, assigning access rights, and establishing dispute resolution mechanisms. From these empirical observations, she condensed her findings into eight design

principles that characterize long-enduring, self-governing CPR institutions. The first seven principles are relevant for this study: (Ostrom 1990[2011], p. 90-102).

While these principles were developed in more formalised systems, their applicability in non-hierarchical communities remains an open question explored in this thesis.

1. **Clearly Defined Boundaries**

Both the resource boundaries and the community of authorised users must be clearly defined. This prevents ´outsiders´ from harvesting benefits without contributing to maintenance and prevents the *tragedy of the commons* from occurring.

2. **Congruence Between Rules and Local Conditions**

Appropriation and provision rules must fit the specific ecological, social, and economic conditions of each setting. Without a one single solution, rules must be tailored to local conditions.

3. **Collective-Choice Arrangements**

Those affected by the rules can participate in modifying them. Participatory decision-making increases legitimacy and compliance because people follow rules they helped create.

4. **Monitoring**

Community members watch over their own water system and each other's behaviour. This creates trust and accountability from within, without needing external inspectors.

5. **Graduated Sanctions**

Punishments start small and get bigger if someone keeps breaking the rules. A first mistake gets a gentle reminder; repeated problems get stronger consequences.

6. **Conflict-Resolution Mechanisms**

The community needs easy and affordable ways to resolve disagreements about water use. When conflicts can be addressed quickly and locally, small disputes don't escalate into bigger problems that threaten the entire system.

7. **Minimal Recognition of Rights to Organise**

External authorities must acknowledge the community's right to create and manage its own rules. If external powers don't recognize this right, they can override or invalidate local decisions, making self-governance impossible.

3.3.1 Application of Design Principles to the Catalan Collective

The rural affinity collective in Catalonia already demonstrates three of Ostrom's seven design principles, while four require deliberate implementation for the RWHS.

3.3.2 Principles Currently in Place

Principle 1: Clearly Defined Boundaries: The collective has an established membership and territorial boundaries.

Principle 3: Collective-Choice Arrangements: Participatory decision-making is embedded in the collective's governance culture. With biweekly meetings, clear communication between all members in chat-forums. Members can collectively determine water access rules, maintenance schedules, and system modifications, creating ownership and commitment to shared decisions.

Principle 6: Conflict-Resolution Mechanisms: The community has assemblies, designated team regarding both water and safety. As well as having recurring informal conflict resolution.

3.3.3 Principles Requiring Implementation

Principle 2: Congruence Between Rules and Local Conditions: Rules must address Catalonia's seasonal rainfall patterns, diverse water needs (drinking, household, agricultural), technical specifications, and proportional contributions based on household size and usage and adapting strategies during periods of droughts.

Principle 4: Monitoring: Considering the collectives non-hierarchical structure, both monitoring and the following *Principle 5. Graduated sanctions* are difficult to implement. Community members would monitor water quality, system maintenance, storage levels. However, denying anyone access to water or adding a system of fines will not be possible. Monitoring approaches should align with existing social structures rather than external oversight. (However, locating the central tap of the systems in the courtyards, adds a prospect of being exposed, and thus raising the stakes of breaking norms.)

Principle 5: Graduated Sanctions: A restorative framework should address rule violations. From gentle reminders for first-time mistakes to stronger measures such as exposure. Reflecting the collective's solidarity values and respect of local norms.

Principle 7: Minimal Recognition of Rights to Organise: Spanish authorities currently tolerate the collective's self-governance, but this recognition is informal. To protect the collective's right to manage its own rainwater system, the community also must have the right to dwell in their current homes.

Importance for research

Ostrom's framework offers this thesis a practical way to understand RWH as a commons rather than only a technical system. The Catalan affinity collective already shows a strong basis for shared governance through clear membership boundaries, participatory decision-making, and existing conflict-resolution practices, which makes it a promising case for collective water management.

At the same time, the framework also reveals where adaptation is needed. Rules must fit Catalonia's seasonal rainfall patterns and the collective's specific water needs. While monitoring and accountability must work within a non-hierarchical structure. Because formal fines or exclusion are not compatible with the group's values, these principles need to be translated into the affinity group's social structure. How can monitoring align with non-hierarchical values? How can accountability function without a system of sanctions? These questions cannot be answered through technical design alone but require the participatory, ethnographic approach central to this research.

This makes Ostrom's design principles especially useful for the thesis, since they turn an abstract commons theory into concrete questions about how the RWHS can be governed in practice. Rather than assuming one universal model, the framework helps identify what is already in place, what needs to be developed, and how rainwater can become a shared resource that fits the collective's social organisation.

4 CASE AND CONTEXT

4.1 CASE: THE AFFINITY COLLECTIVE

4.1.1 Social Structure

The affinity group is a non-hierarchical queer collective. Which consists of approximately 50 individuals, ranging in age from 2 to 60, who inhabit four off-grid buildings entirely disconnected from municipal infrastructure. The site covers about a hectare, and rooftop area is 200m^2 for each building. While residents live in a mixed living constellations, the community operates on an established cultural norm of "togetherness in moderation." Common spaces are highly adaptable and multi-functional: a gymnasium doubles as a children's playground, and a room used for raves on the weekends frequently hosts quiet boardgame nights during the week.

Spatially the boundaries between private and common areas are highly fluid. The sense of space extends beyond the walls. Windows act as conversational thresholds to the courtyards outside. And communication does not rely on phones, but by knocking or shouting.

Socially, the collective practices selective gatekeeping, a necessity both for physical protection and for maintaining group cohesion. Founded by a group of trans people, the community has grown into a sanctuary for many, described by residents as "*the place in Spain with the most trans people per square meter*." Today, it hosts a highly diverse, international population from countries such as Ukraine, Colombia, and Argentina.

Recently, the community has gained visibility in queer and leftist media, resulting in a growing waitlist of people hoping to move in. While this popularity reflects the success of their living model, it has also introduced new social and logistical pressures. To manage this growing complexity, during my second site visit, the collective began decentralising its governance, forming focus-groups to handle specific operational areas, such as water, electricity, safety, events, legal & publicity.

While the collective demonstrates strong cohesion and shared values, fieldwork also revealed emerging challenges related to scale, coordination, and uneven distribution of responsibilities.



Figure 7: Building with its many entrances and public patio. Analogue photograph by author.



Figure 8: Against the Mountain. Analogue photograph by author.



Figure 9: Green Anarchy. Digital photograph by author.



Figure 10: Social gathering in the "avenida" Digital photograph by author.

4.1.2 Economic Framework

The collective presents a unique case where standard economic models are not applicable. Proposing to pay anything instead of nothing creates a dilemma for the community: who is responsible, and how are the costs divided among members? This is especially challenging in a community based on favours and the exchange of goods rather than money.

For instance, Net Present Value calculations, as utilised by Domènech and Saurí (2011), willingness to pay (WTP) analyses, such as Contingent Valuation Method and Revealed Preference Method employed by Mukarram et al. (2023) are not applicable to the group's financial reality. While previous studies rely on these frameworks to measure the economic value of communal water systems, this affinity group operates completely outside traditional market dynamics. Consequently, such models were excluded, as their underlying market logic fails to capture the collective's non-hierarchical and non-monetary modes of organising and valuing shared resources.

The economic structure shapes the project, since conventional valuation methods do not reflect how the group organises shared resources. This reinforces the need to understand RWH through social practice rather than market logic.

4.2 CURRENT WATER SYSTEMS

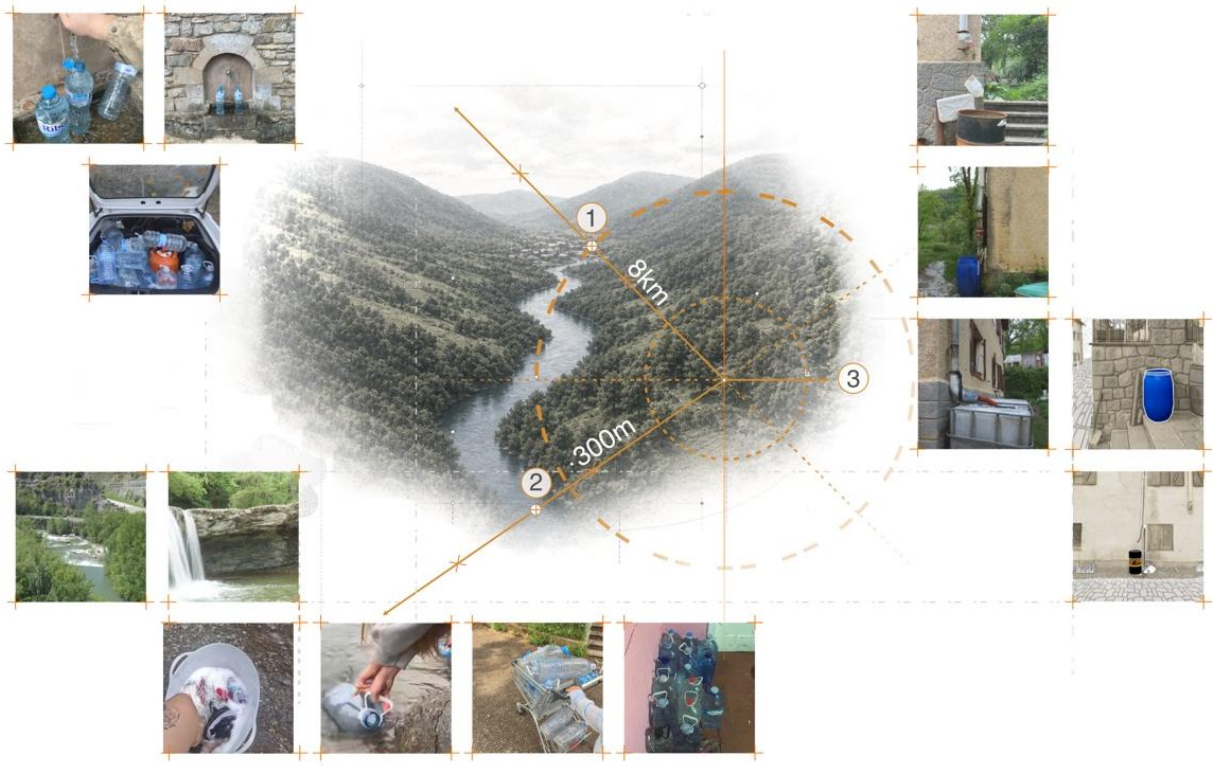


Figure 11: Collage of Current Water Systems / Sources used. All photographs by author.

Water is collected from three sources: public fountains in nearby villages, the river downhill, and rainwater harvesting stored in barrels. The first source provides potable water and is transported to the site from about 8 km away, usually by shared car or public transport. While water collection is often handled individually, collaboration is common, particularly when transport can be combined with other trips such as school or work. The latter two sources are collected depending on accessibility and are shared in the kitchens for washing and in the bathrooms for flushing and sanitation. however, river collection is labour-intensive, requiring long walks with heavy lifting and hand submersion into cold water, particularly challenging in the winters. Both sanitary procedures and washing clothes are practiced on the site and by the river.

4.3 SITE CONTEXT AND CLIMATE

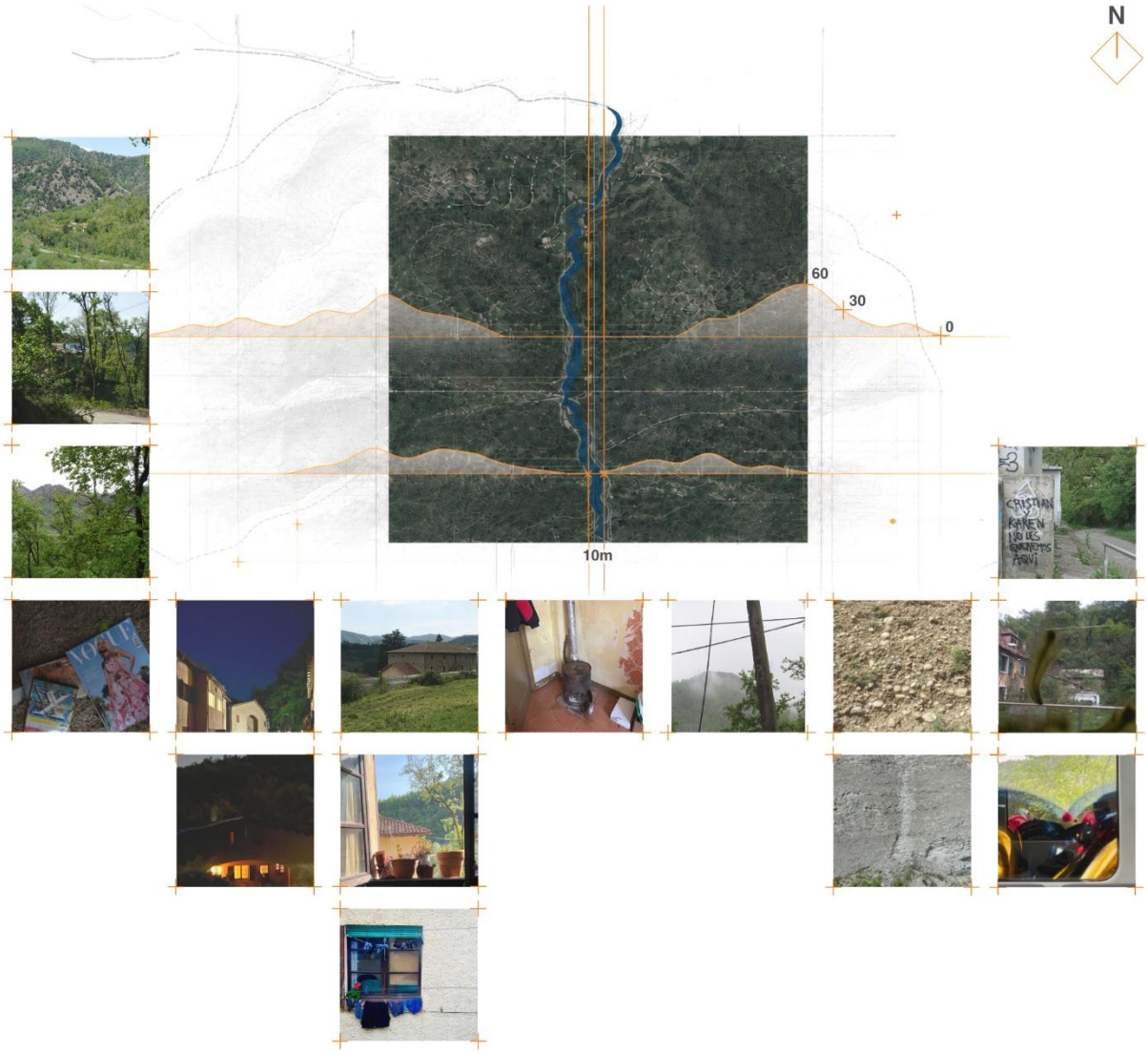


Figure 12: Collage of the Site. All photographs by author and the "satellite image" is AI-generated using described characteristics of the site.

This research is situated in rural Catalonia, Spain, within a Mediterranean climate marked by seasonal variability, irregular precipitation, prolonged dry periods, and recurring water stress. To protect the anonymity of the collective, the exact site is not presented. Instead, the study presents a generalised rural and mountainous context that reflects the environmental conditions of the site.

4.3.1 Precipitation

Rainfall data is key to determining if RWH is feasible on a specific site. Rainfall varies widely across Catalonia. However, the site gets more rainfall than most areas, receiving around 900 millimetres of rain per year (Idescat, 2025), showing good potential for RWH. Rain falls unevenly through the year, thus monthly data helps assess seasonal supply and storage needs. (Calculating storage requirements explained in Appendix A: Storage Requirements and Water Balance Analysis)

This potential must be understood within a context of increasing climate-instability. In 2024, Catalonia experienced a severe drought emergency that affected more than six million people and led to water restrictions of 200 litres per person per day, while reservoir levels fell to 27% (Brown & Formanek 2024). These conditions underline the importance of decentralised water systems and support the relevance of RWH as a strategy for local water resilience. High feasibility of RWH has also been proven in local context by Domènech, March and Saurí (2013).

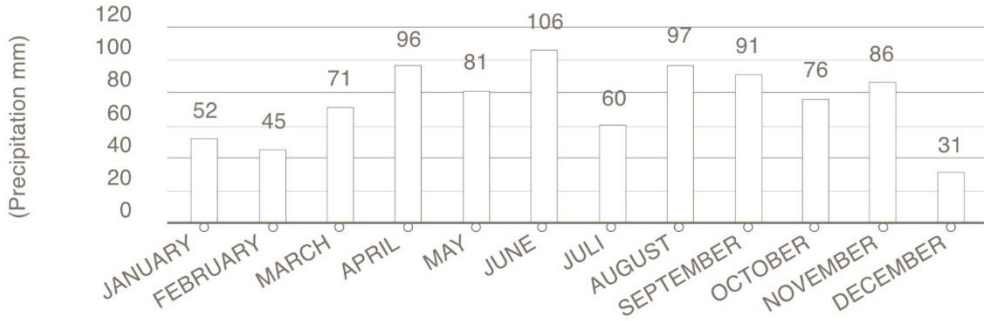


Figure 14: Average Monthly Precipitation (mm) from 2014-2024. Created by author with data from Idescat (2025).

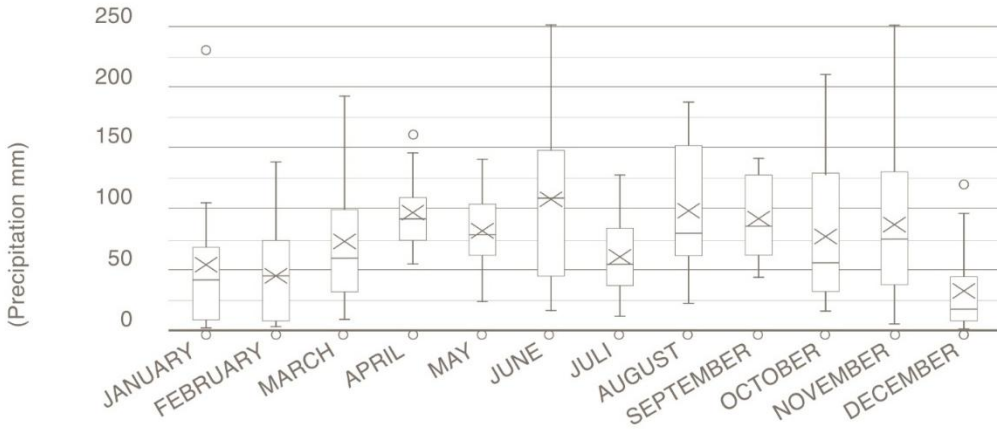


Figure 13: Monthly Precipitation (mm) demonstrating variation from year to year between 2014-2024. Created by author with data from Idescat (2025).

5 INITIAL DESIGN PROPOSAL



Figure 15: Vision of BSFS as part of Proposal. Drawing by author.

5.1 PROPOSED BIOSAND FILTER

5.1.1 Rationale for Selecting Biosand Filtration

The initial selection of Biosand filtration (BSF) for the collective's potable water system was chosen because of its alignment with the community's off-grid context and goal of autonomy. Unlike other modern technical filtration methods such as reverse osmosis and ultraviolet filtration, which require consistent electrical power and filter replacement, BSF is a low-tech, gravity-fed system that is vernacularly anchored and uses natural components. From a socio-material perspective, this technology is highly appropriate for an affinity group that manages water collectively on-site. It can be constructed using locally accessible and inexpensive materials and maintained directly by the community's newly formed Water group. By not depending on external supply chains or energy consumption, BSF supports infrastructural self-sufficiency. However, this proposal also assumed a level of collective responsibility that had not yet been verified socially.

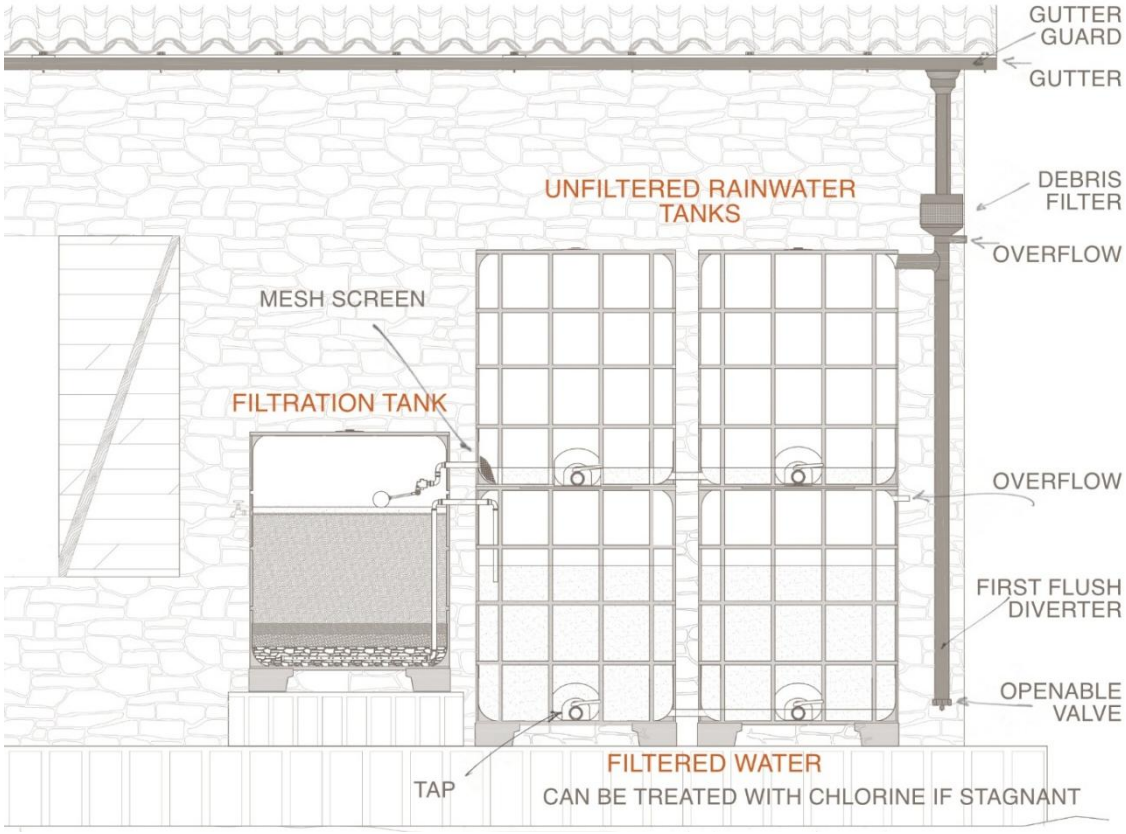


Figure 16: Biosand Filtration System and Components. Drawing by author.

5.1.2 Technical Description

A Biosand filtration system (BSFS) is a household water treatment, using traditional Slow Sand Filtration (SSF). Consisting of a concrete or plastic container, layered with filter media (gravel and sand). Above the sand there is a shallow level of water (~5cm), allowing for a biological layer (Schmutzdecke) to develop. The layer contributes to removing pathogens by competing out harmful organisms in the water. SSF are highly effective for RWH, using gravity flow through layered sand to remove suspended sediments. (Dangol & Spuhler n.d)

Multiple mechanisms work together: (1) Mechanical filtration traps large particles between sand grains. (2) Biological metabolism uses created bacteria colonies on sand surfaces (Biofilm), which breaks down organics. (3) Electric changes neutralise charged impurities via opposite charges on sand particles. This reduces bacteria significantly and alters water chemistry. (Joji & Jacob 2023, p. 93-94)

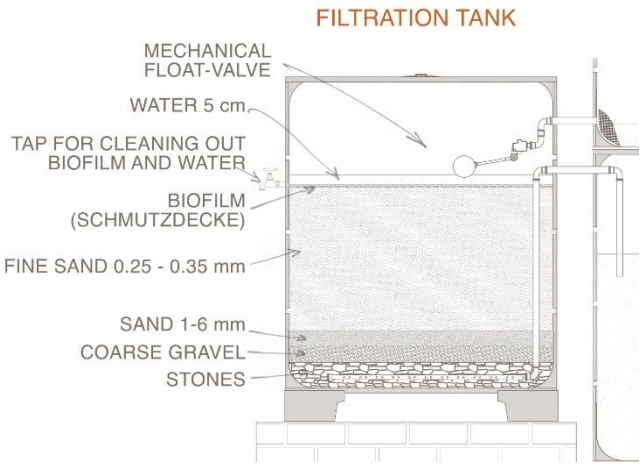


Figure 17: Detailed Drawing of Filtration Tank. Drawing by author.

5.1.3 Water Balance for Potable Use

A water balance analysis is a practical accounting tool used to figure out if a RWHS will work for a specific site. It compares the incoming water supply (based on local rainfall and roof size) against the community's everyday water use. By tracking these numbers month by month, the analysis reveals exactly when the collective will have surplus water and when they might face a deficit. For an off-grid community aiming for self-sufficiency, this calculation is crucial. It proves whether year-round water autonomy is genuinely feasible and ensures the storage tanks aren't built too big or too small. (See appendix A: Storage Requirements and Water Balance Analysis for full technical explanation). Water balance for the BSFS regards the collectives 50 members and a consumption of 7 L/c/d (Litres per capita per day).

Water demand:

$$\mathbf{10500\ L}\ (10.5m^3)\ =\ 50\ x\ 7\ x\ 30$$

Water supply:

$$\mathbf{13.500\ L}\ (13.5m^3) = 200m2\ x\ (\frac{900mm}{12})\ x\ 0.9$$

Storage requirement:

$$4.000L\ (4m^3)\ =\ 50\ x\ 7\ x\ D$$

Calculating backwards for the storage requirement from knowing the tank size, gives, D (days of potential drought) ≈ 11 days. While a daily consumption of 3 L/c/d from a full tank would last four weeks.

5.1.4 Rainfall Analysis

As shown in Figure 14, rainfall over the past ten years varies considerably, with some years, January receiving almost no precipitation and others receiving more than 100mm. This variability indicates that a system relying only on rainfall would be difficult to sustain for a group of this size, as it would require very large storage capacity to bridge dry periods.

Figure 13 shows the average monthly precipitation over the last ten years, which can be compared to the estimated monthly water demand of approximately 60 mm of rainfall, or 10.5 m^3.

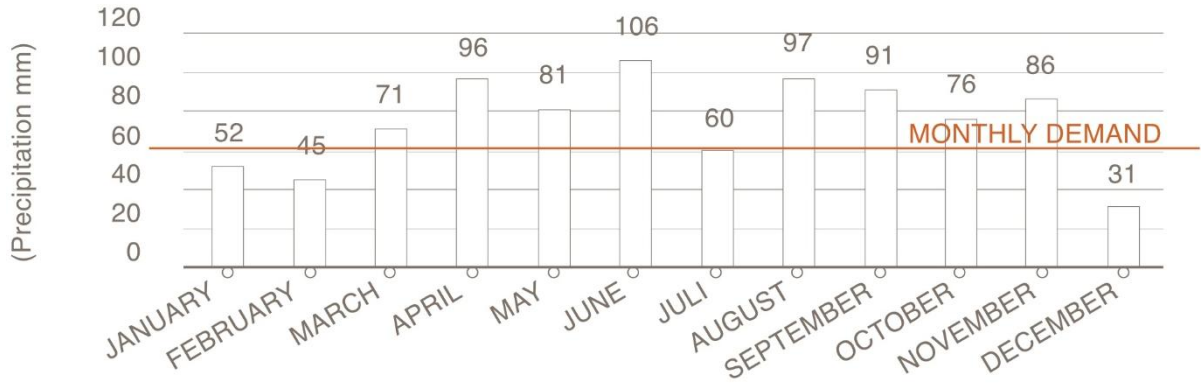


Figure 18: Average Monthly Precipitation and Monthly Demand.

As shown in Figure 18, average monthly precipitation appears sufficient to meet monthly demand, but seasonal variation means that some months still require storage and careful demand management. In particular, the model suggests that tanks may run dry during winter, indicating that RWH cannot function as a single potable source. The results therefore point to the need for flexible strategies, either by maintaining the current water system or by developing a shared storage model that distributes water more effectively across the collective.

5.1.5 Adaptive Demand Strategies

To address Catalonia's seasonal rainfall variability, the proposed RWHS incorporates tiered demand strategies based on real-time storage levels. Visual indicators on tanks would signal three operational zones: green (>75% capacity), yellow (50-75%), and red (<50%). In green zone conditions, normal use continues for all purposes. Such as washing dishes, cooking and drinking. Yellow zone conditions trigger prioritisation of potable needs. Red zone conditions require strict rationing, drinking only. With the collective reverting to water fetching as currently used for all other needs. Where the red zone would lower daily consumption to 3L/c/d.

If the rainwater tank capacity drops below 25%, the community can use an additional strategy by dividing members into two groups. One group includes people who have more difficulty collecting potable water, for example due to health reasons or families with children. Allowing only this group to continue using the local filtered water, while the rest of the community collects drinkable water using the same method they currently use.

5.1.6 Maintenance of RWHS and BSFS

Regular maintenance of all components is required for optimal harvesting and quality of rainwater. First flush systems should be emptied after each storm. Gutters should be clean from debris. All piping and connections checked for good function. Filters and cisterns must be checked and cleaned periodically (Kinkade-Levario 2007, p. 32) and all screens on inlets and outlets needs screens to prevent mosquito larvae.

Rainwater for potable use should regularly be tested for quality. The quality is dependent on the catchment surface, gutter and downspout material. Thus, an initial water quality test must be done to evaluate components. As well as every six months, minimum. Water quality should be tested at least for turbidity and E. Coli, however many parameters (such as alkalinity and PH) can be examined depending on budget and overall quality of the RWHS.

The types of contaminants usually found in RWHS are listed by SOPAC Water quality officer Mosley (2005, p. 4-8) as: Suspended sediments, pathogenic bacteria, heavy metals & chemicals and mosquito larvae. (Cause and prevention are further explained in Appendix B: Rainwater Contaminants).

REFRAMING THE INTERVENTION

The initial proposal for a Biosand filter was developed as a robust, low-energy solution for on-site water treatment. However, presenting it to the collective revealed a sharp disconnect between technical feasibility and social reality. Residents' hesitations concerned both the filter's performance, its maintenance, but above all the responsibility of a selected few, overseeing the health of many.

Rather than a failed design outcome, this rejection served as a crucial empirical finding: RWH infrastructure regarding potable use, cannot be implemented without trust, cannot be managed without confidence, and the governance must be of a bigger group to function. This prompted a methodological pivot away from a purely technical installation toward a participatory process.

The intervention evolved into a collaborative, artistic transformation of the infrastructure. By painting the water tanks together, which practically blocked UV light to prevent algae growth, the community visually and culturally reclaimed the tanks. This shifted the project's focus from maximizing technical efficiency to nurturing social engagement.

In doing so, the project moved from delivering a predefined "solution" to creating a space with the collective where future water practices could be explored.

6 RESULTS

6.1 TECHNICAL FEASIBILITY OF RAINWATER HARVESTING

The results show that historical Mediterranean RWH strategies can be adapted to the collective through simple, low-tech components such as roof catchment, storage tanks, and adaptive demand strategies. Rooftop RWH is especially suitable because of the site's large catchment area, existing gutters and drainage pipes, and space for expandable storage.

This suggests that, in relation to the first research question (1) RWH can be introduced by upgrading and connecting existing structures rather than by implementing high-tech systems.

Water balance analysis indicates strong technical potential in the rural Catalan context, where average annual precipitation is approximately 900 mm. When water use is limited to drinking and cooking, at about 7 L/capita/day, rooftop RWH can theoretically meet monthly potable demand in average rainfall years with medium-sized storage tanks. During drought periods, adaptive demand strategies that reduce consumption to around 3 L/capita/day provide roughly four weeks of potable self-sufficiency. These findings show that RWH can materially support self-sufficiency, but only if it is combined with flexible, tiered demand that responds to changing storage levels.

At the same time, the water balance analysis confirms that RWH cannot function as a single water source, neither for potable nor non-potable use. Seasonal variability requires sufficient storage, careful use, and a diversified set of water sources. In practice, the collective must maintain current water collection methods while shifting more everyday uses toward rainwater stored on site. In relation to the first research question (1), the technical findings suggest that RWH is best introduced as a complementary system that reduces vulnerability and travel time, rather than as a total replacement.

Within this broader framework, the water balance analysis for the BSFS was based on a single building rooftop area of 200 m². This approach reflects both the limited material resources available and the intention to present a pilot system using five new IBC containers, designed only to supply potable water. A BSFS also proved technically efficient, with estimated costs of 700€, and was conceived as a modular intervention targeting the most distant water source. In future applications, extending the system to multiple buildings would significantly increase yield and improve year-round accessibility. With strategic storage, such an expanded system could potentially provide sufficient potable water for the entire collective of 50 members.

However, the rejection of the BSF directly links to the second research question (2). T*echnical feasibility alone is insufficient if governance, trust, and responsibility-sharing are not aligned*. Overall, the technical results show that rooftop RWH can substantially support self-sufficiency, provided storage is sized to local climate, demand remains adaptive, and RWH is integrated as one element within a collectively governed mix of water sources.

6.2 SOCIAL ACCEPTANCE AND GOVERNANCE CONSTRAINTS

Despite its technical feasibility, the proposed BSFS encountered significant scepticism from the collective and was ultimately rejected. Primary concerns centred on uncertainty about water quality, the high costs of regular testing, and, above all, the risk that responsibilities would fall on a small number of people in a non-hierarchical group of 50, being responsible for the health of all, including vulnerable users such as children. This rejection exposed a key socio-technical barrier directly relevant to both research questions: potable water infrastructure in off-grid collectives requires not only a robust design, but also collective trust, shared responsibility, and governance arrangements that the group perceives as fair and manageable.

Existing studies reflect similar patterns of cautious acceptance. In Algeria, households were more willing to adopt RWH when government subsidies reduced economic risk (Bouzina, Belmeziti & Gouvelle 2024). In Sabadell, Catalonia, residents viewed RWH as technically promising but expressed worries about maintenance, water quality, and the absence of visible, functioning examples in their neighbourhoods (Domènech, March, & Saurí 2013). Mukarram (2023) further observed that in cases when people can see RWH working nearby, adoption rates increased fourfold, thanks to hands-on experience and trust built through word-of-mouth.

In the Catalan collective, the rejection of the BSF demonstrates that, in practice, *low-tech solutions may generate greater anxiety precisely because responsibility cannot be delegated to external authorities or professionals*. The collective's response shows that any future potable RWH intervention must be co-designed to distribute responsibility more evenly, ensure that water quality is legible and trusted, and embed governance mechanisms that fit the group's non-hierarchical structure.

In terms of the second research question (2), this episode reveals that governing rainwater as a commons depends on more than physical access. *It depends on whether the community feels able and willing to collectively carry the risks associated with managing drinking water*.

6.3 PARTICIPATORY PIVOT AND INFRASTRUCTURE APPROPRIATION

Following the rejection of the BSFS, the project pivoted to a participatory intervention that repurposed the purchased transparent water tanks. Over four workshops across a two-week period, complemented by spontaneous community efforts, participants collaboratively determined tank placement, and later added graffiti artwork. On a technical level, participants painted the tanks black for UV protection, prepared and installed the interconnected PVC-piping system. The later maintains equal water levels across all tanks, preventing individual overflow and maximising total harvest capacity. On a social level, the activities encouraged broad community ownership, with 10–25 percent of residents taking part in different moments. The more technically oriented work mainly involved the water focus group, while the collective

graffiti artwork drew in a wider demographic of children, parents, and young adults who embedded personal meanings into the infrastructure. (For a detailed breakdown of the workshop phases, participant demographics, and specific community artwork, see Appendix C: Workshops and Participation & Appendix D: Final Outcome of Artwork and Qoutes)

Participants reported feeling more connected to the infrastructure after the workshops, and informal conversations during the process helped shape emerging norms around use and responsibility. Community members now actively use the expanded system for non-potable needs such as washing and gardening, and several noted that the *visible, painted infrastructure made water feel like a shared resource rather than something belonging only to those who participated in the workshops*.

The participatory tank-painting workshops generated visible engagement that community members described seeing the RWHS as 'their' resource, rather than an external supply. *Creating communal landmarks as a cultural expression*.

These developments align with the collective's non-hierarchical structure, where participatory processes enabled the formation of shared rules without formal enforcement mechanisms. Taken together, the findings suggest that RWH in this context functions as a socio-material practice in which infrastructure, everyday routines, governance, and cultural expression develop together.



Figure 19: Workshop 3. Participatory art. Further explained in Appendix C: Workshops and Participation. Collage and Photographs by author.

6.4 BUILT RAINWATER HARVESTING SYSTEM



Figure 20: RWHS: Product of Workshops.
Photographs by author.

7 DISCUSSION

7.1 DESIGN FAILURE AS PRODUCTIVE KNOWLEDGE

The rejection of the BSFS illustrates how technically sound design proposals can fail when disconnected from social realities and fail to gain social acceptance. Rather than undermining the project, this failure produced critical insight into how off-grid water systems are governed, trusted, and maintained. The case reveals that low-tech solutions may paradoxically generate greater anxiety precisely because responsibility cannot be delegated to external authorities or professionalised systems. In contrast, high-tech filtration systems, would hold the producers of the filtration system equally accountable.

The community's rejection of the system demonstrates that off-grid water infrastructure *cannot be successfully implemented if health responsibilities fall on a certain few.* This rejection highlighted a complex local dilemma regarding water safety. While the community places high trust in advanced, highly technical filtration systems, such solutions remain financially out of reach. Conversely, because affordable off-grid adaptations must prioritise simplicity, the community expressed scepticism and low trust toward ´simple´ low-tech water treatment methods.

In this context, the social failure of the Biosand proposal was deeply productive. It shifted the focus of the research, reinforcing the understanding that RWH as a socio-material practice. Viewing RWH not as a static technical system, but an ongoing, embodied, and collective performance in which social relations and material infrastructure must continually co-evolve, and *where knowledge by the user is more important than knowledge by the builder*. By pivoting away from an unwanted filtration system and adapting to the community's actual comfort levels, the project demonstrated that art and design is not just an aesthetic effort, but a tool to encourage participation. In creating and building the existing RWHS, the community gained knowledge in water quality, preservation methods and construction, ultimately reinforcing human resilience in an unstable environment and opening possibilities for continuous construction.

For the collective, next steps include gradually expanding catchment and storage, co-developing monitoring tools and maintenance rituals, and returning to potable filtration only when governance structures feel ready to carry the associated responsibilities.

7.2 GOVERNING RAINWATER AS A COMMONS

Drawing on Ostrom’s theory of CPRs, the study shows that rainwater functions as a shared resource and that its sustainability depends on collectively negotiated rules, monitoring, and participation. In this case, rainwater was governed not through formal regulation, but through shared practices of visibility, care, and appropriation. The artistic intervention supported a non-

institutional form of commons governance by making the infrastructure socially legible and symbolically meaningful.

Many factors suggest that the collective is well positioned to maintain the RWHS as a commons (as reviewed in 3.3.2 Principles Currently in Place). Water is already treated as a shared resource through the collection of water from open sources and the small-scale harnessing of rainwater, even though the current system has vulnerabilities. The community already fulfils three of Ostrom's design principles: *clearly defined boundaries, collective-choice arrangements, and conflict-resolution mechanisms*. In addition, their relatively closed setting, with close ties and routines around water, supports the development of strong norms of acceptable behaviour. This echoes Ostrom's observations from cases where people live close to their CPRs and therefore build everyday practices of care and responsibility (Ostrom 1990[2011], p. 206).

7.2.1 Adapting Ostrom's Principles to Non-Hierarchical Governance

The collective's non-hierarchical structure creates tensions with Ostrom's design principles, particularly regarding monitoring and accountability. *The community rejected the BSF partly because health responsibilities fell on a certain few, revealing deep resistance to authoritarian roles*. Traditional monitoring systems assume designated authorities, yet the collective operates without hierarchy. Similarly, the principle of graduated sanctions seems incompatible with a community where denying anyone access to water or adding a system of fines will not be possible. These contradictions are not dead ends. However, they open chances to rethink and improve how systems are organised.

Monitoring can align with non-hierarchical values when reframed as mutual care rather than surveillance. Locating the central tap of the systems in the courtyards adds a prospect of being exposed and thus raises the stakes of breaking norms. This spatial design creates passive monitoring. Water use becomes naturally monitored without requiring designated watchers.

Accountability can function without punitive sanctions when the collective adopts restorative approaches. A restorative framework addresses rule violations through graduated responses. However, Ostrom also presents how *shared norms lowers need of monitoring* and "*sanctioning activities can be viewed as social capital to be utilized in solving CPR problems"* (Ostrom 1990[2011], p. 36)

In this context, adaptive demand strategies, such as lowering consumption when storage levels decline, help distribute scarcity more evenly and prevent tension between those who follow the rules and those who do not. Annual maintenance workshops, including tank cleaning, gutter maintenance, and collective celebrations when the tanks are full, can further turn care into a shared ritual and make compliance something that emerges from within the community rather than being imposed from outside.

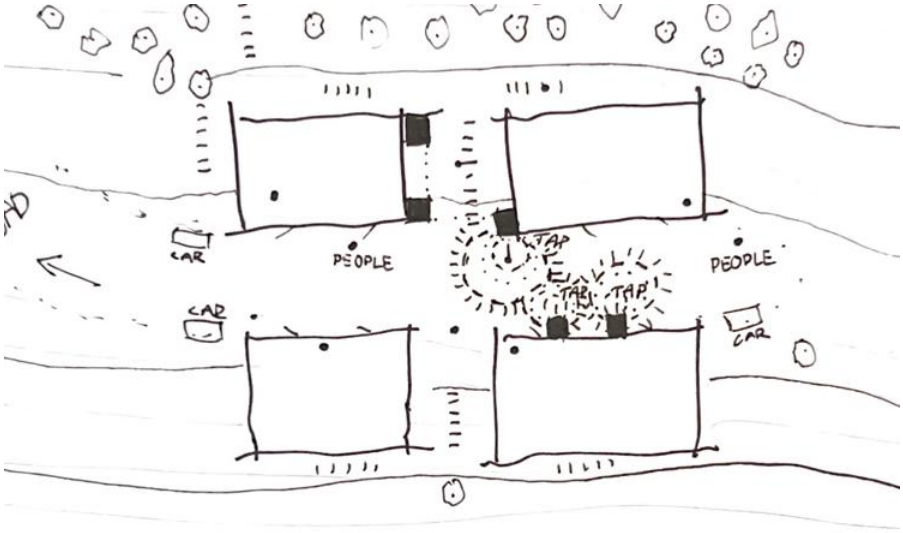


Figure 21: Deciding Location of Tanks
in Relation to PRA-Framework.

7.3 RAINWATER HARVESTING AS A SOCIO-MATERIAL COMMONS

The concept of commons, traditionally rooted in natural resource management, must be expanded when applied to off-grid non-hierarchical communities. In the context of the Affinity group, RWH goes beyond the mere management of an existing natural resources. It is a practice actively created through human intervention. Rainwater is abundant yet irregular, and it only becomes a reliable resource when captured, stored, and governed by the community. *Rain as a commons* reframes rainwater into a commons-oriented way of dwelling.

Implementing a CBRWHS removes water from municipal or profit-driven systems, restoring direct, democratic governance over the resource. Addressing how rainwater can be collectively governed to enhance shared responsibility, reveals that *governance in a non-hierarchical group relies heavily on cultural and visual integration*. Following Ostrom's principles, governing a commons requires clear boundaries, shared rules, and active participation. In the absence of formal state institutions this collective achieved governance through participatory action, but above all shared norms and lifestyles.

7.4 VISABLE DESIGN AND ARCHITECTURAL EXPRESSION

The community's artistic appropriation of the tanks functioned as an act of collective governance rather than decoration. By transforming mass-produced industrial objects into shared artefacts of identity and memory, the affinity group strengthened a sense of ownership that supports care, maintenance, and long-term adaptation. Art therefore played a central role in establishing RWH as a socio-material practice, merging technical infrastructure with the collective's existing norms of expression and participation.

This approach also challenges the architectural tendency to hide technical systems and separate infrastructure from everyday life. By keeping the tanks visible, the intervention made the water system legible, shared, and socially meaningful. In this sense, visibility was not only an aesthetic choice, but also a governance strategy that embedded the RWHS in the collective's daily spatial and social routines, allowing users to walk past, sit beside, and directly observe water levels rising and falling over time.



Figure 22: Collective Expression.
Analogue photograph by author.

7.5 LIMITATIONS OF THE STUDY

This thesis is limited by the short duration of fieldwork, which consisted of only two site visits over a relatively brief period. As a result, the findings reflect a snapshot of the collective's practices rather than long-term patterns of water use, governance, or maintenance. The study is also based on a single case, which represents a specific social identity and way of living that differs from many other off-grid contexts. In addition, the researcher's position as an outside participant may have influenced interpretation, despite efforts toward reflexivity and ethical care.

These limitations are especially relevant to the thesis's governance questions. While the study suggests that monitoring and accountability must be rethought within a non-hierarchical collective, it cannot show how these arrangements would function over time in practice.

Questions about how monitoring can align with non-hierarchical values, or how accountability can work without sanctions, remain only partly answerable within this research and call for a longer-term study.

CONCLUSION

This thesis showed that RWH in the Catalan affinity collective is not only a technical issue, but a socio-material practice shaped by governance, trust, and everyday life. Framed through Water as a commons, the study demonstrated that RWH becomes viable when technical design is matched with local routines, shared responsibility, and forms of accountability that fit a non-hierarchical community.

The rejection of the BSFS shows that even technically sound systems fail when responsibility of all, falls on a certain few. Which differs from previous studies, where concerns centred on water quality or economic cost. Which proves that potable systems are accepted only when trust, responsibility, and risk are distributed in ways that feel fair and manageable within the collective's structure. The participatory pivot that followed proved more productive.

The findings answer the research questions in two ways. First, rooftop RWH can support self-sufficiency through low-tech upgrades, adaptive demand, and shared storage, but only as part of a diversified set of water sources rather than as a single source. Second, rainwater can be governed as a commons when it is visible, culturally integrated, and managed through mutual care instead of fixed hierarchy. Ostrom's design principles, especially monitoring and sanctions, could not be transferred directly to the non-hierarchical structure of the collective and instead had to be rethought through visibility, mutual care, and adaptive practice.

In this sense, the project challenges modern architectural ideals that separate water from everyday life and hiding technical systems from view. Further emphasising the combination of visible infrastructure and art as a practice of participation and local cultural expression, enhancing self-sufficiency and collective responsibility.

The main contribution of the thesis is to reframe rainwater harvesting in the Global North as a socio-material commons, where architecture, sustainable living, and collective governance develop together.

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9 APPENDIX

A. STORAGE REQUIREMENTS AND WATER BALANCE ANALYSIS

STORAGE REQUIREMENT:

The size of the water tank is calculated in relation Water balance and estimated periods of water scarcity. (Joji & Jacob, 2023, p. 98)

Size of Water Tank:

- N – Occupants
- L/c/d – Daily consumption
- D – Period of drought (days)

$$Size\ of\ storage\ tank\ =\ N\ x\ (L/c/d)\ x\ D$$

WATER BALANCE:

A Water budget (Water Balance Analysis) compares monthly rainwater supply against water demand. Revealing storage needs and periods of water scarcity. (Kinkade-Levario 2007, p. 36-41)

Daily water demand varies widely across the globe. Ranging from 10 Litres / capita / day (L/c/d) in several parts of the world to several thousand L/c/d in some developed countries (Joji & Jacob, 2023, p. 100) Rahman presents different levels of water demand. Presenting *Basic access*: 20 L/c/d as basic access- meeting water demands of consumption and hygiene. *Intermediate access*: 50 L/c/d. Also meeting food preparation and *Optimal access*, at 100 L/c/d. Meeting all hygiene and consumption needs. (Rahman 2018, p. 114)

Water Demand:

- Wdemand (m^3) - Water demand
- N – Occupants
- L/c/d - Daily consumption litres per capita per day
- 30 or 365 for monthly / annual consumption

$$Wdemand\ =\ N\ x\ (L/c/d)\ x\ (30\ or\ 365)$$

Water Supply:

- Wsupply – Water supply (m^3)
- A – Roof size (m^2)
- P – Monthly or Annual Precipitation (mm)
- C – Runoff coefficient

$$Wsupply = A \; x \; P \; x \; C$$

RUNOFF COEFFICIENT:

For a roof to yield as much water as possible. Roofing materials should have impermeable surfaces, free from contaminants and good connection to guttering. The runoff coefficient. Is depending on type of roofing.

ROOFING MATERIAL	RUNOFF COEFFICIENT
Galvanised iron sheets	>0.9
Glazed tiles	0.6 - 0.9
Asbestos sheets	0.8 – 0.9
Gravel roofs, paved surfaces	0.8
Concrete	0.7
Treated soil	0.6
Natural soil	0.3
Thatched- or palm- roofs	0.2

(Thomas & Martinson 2007, p. 41) (Joji & Jacob 2023, p. 98) (Kinkade-Levario 2007, p. 38)

B. RAINWATER CONTAMINANTS

SOPAC Water quality officer Mosley lists contaminants usually found in rainwater harvesting systems (2005, p. 4-8) Preventions listed, are before entering the filtration process. Following table also includes information from Thomas & Martinsons chapter on the topic. (2007, p. 39-43).

CONTAMINANTS	CAUSE AND PREVENTION
Suspended Sediments	Suspended sediments are the most common pollutant, washing off roofs as dirt, vegetation or ash. While usually non-toxic, it causes cloudy water (turbidity) and can carry microbes. Fortunately, 90% of SS settles to the tank bottom within 24 hours. **Prevention:** Roof and gutter cleaning, and first-flush diverters.
Pathogenic Bacteria	Bacteria (like E. coli) enter the system via animal droppings on the roof. Bacteria levels spike after rain but naturally drop by 90% within three days due to settling and die-off. **Prevention:** Regular cleaning of catchment areas and tanks, plus first-flush systems.
Heavy Metals & Chemicals	From nearby urban pollution, industrial emissions, and the roof materials themselves can introduce heavy metals or chemicals. Rainwater is naturally slightly acidic (pH 6), but urban rain can be highly acidic (pH 4–5), which increases the leaching of metals from roofs. Conversely, cement tanks can absorb calcium and raise the pH to 8–9. **Prevention:** Use safe, non-toxic roofing materials, install gutter guards, and use first-flush diverters.
Mosquito Larvae	From mosquitos laying eggs in gutters and/or tanks. **Prevention:** No gaps on construction, mesh screens in every in-and- outlet.

C. WORKSHOPS AND PARTICIPATION

While much of the community's work happened spontaneously through informal walk-and-talks, ´as needed´ tasks, and discussions during other events, the formal participatory research was structured around four planned workshops. These structured sessions took place over four days, split across a two-week period.

With the purpose of engaging, teaching and provide knowledge for future maintenance the workshops showed both great importance of technical installation. As well as social factors, improving the concept of collective ownership over the new infrastructure.

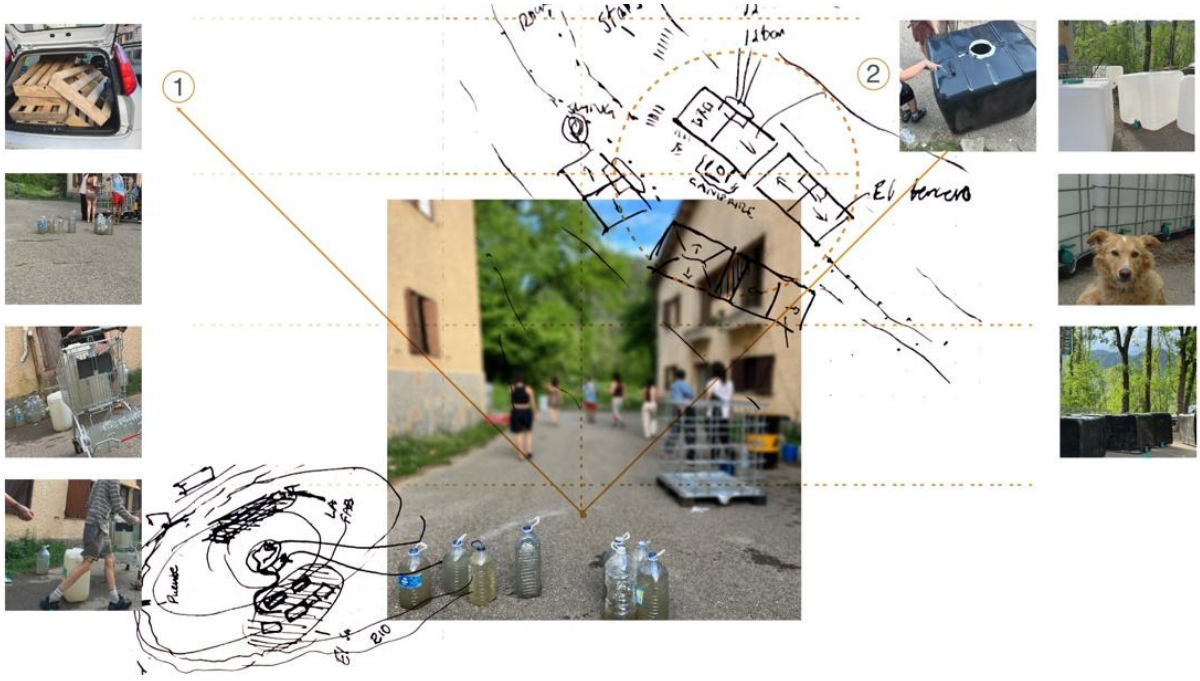


Figure 23: Workshop 1 & 2: Sourcing and Preparing.

WORKSHOP 1 & 2: SOURCING AND PREPARING

The first two workshops focused on sourcing materials, determining the optimal locations for the water tanks, and applying three layers of black paint to the exteriors. Participation was approximately 20% of the collective. This group consisted primarily of young adults who I have engaged with the most, all members of the “water focus group” participated. These workshops were highly engaging, focusing more on letting all opinions heard, but also planning on whom is responsible for which task. As some were dependent on cars.



Figure 24: Workshop 3: Collective Artwork

WORKSHOP 3: COLLECTIVE ARTWORK

Targeting all members of the collective, the third workshop focused on creating collective artwork to decorate the tanks across the site. Using graffiti cans, residents with more experience taught simple techniques to beginners. Participation peaked at approximately 30% of residents, drawing a diverse mix of children, parents, and young adults.

The artwork naturally reflected this demographic variety. Children drew their names, rainbows, hearts, smileys, flowers, and one piece that playfully evolved from a phallic shape into a worm monster. Meanwhile, the young adults focused on patterns and written quotes. Following the workshop, informal conversations provided a space to discuss the specific quotes written on the tanks, their English translations, and their underlying meanings for the participants is presented in Appendix D: Final Outcome of Artwork and Qoutes.

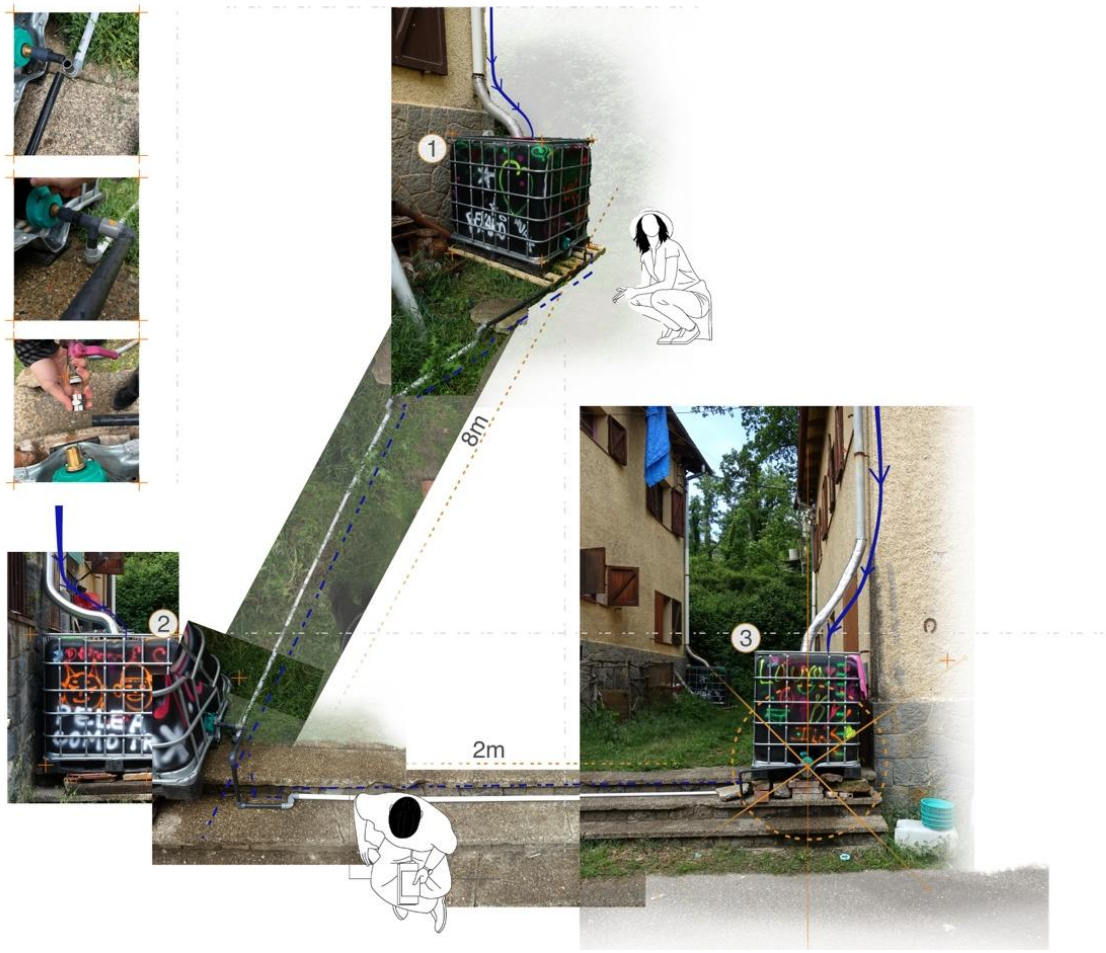


Figure 25: Workshop 4: System Installation

WORKSHOP 4: SYSTEM INSTALLATION

The final workshop engaged the water focus group and other interested residents, representing about 10% of the community. Participants physically installed the tanks and connected them to the roof gutters. We also linked the tanks together using PVC piping to create a single shared outlet. This unified connection ensures all tanks raise and lower in water level simultaneously, eliminating the risk of individual tanks overflowing while maximizing overall harvest capacity and simplifying water storage calculations.

D. FINAL OUTCOME OF ARTWORK AND QOUTES



QOUTES: MEANING AND TRANSLATIONS

“KE OTRXS SEAN LO NORMAL”

(Let others be the normal ones) from the quote "Reivindico mi derecho a ser un monstruo, Que otros sean lo normal" (I claim my right to be a monster, Let others be the normal ones) by Argentinian trans-activist and poet Susy Shock.

“REFORESTA EL CERRO DI TE KIERO”

(Reforest the hill, say I love you) by Trans- & Environmental- activist Emilia Bau, mainly associated to protect indigenous land in Chile. And murdered while participating in a Mapuche land reclamation effort in south Chile in 2021.

“Furia y dolor a mi muerte”

(Rage and pain to my death) “no explanation needed”.

“Love mom”

participant left no comment

“PELEA COMO TRAVA”

(Fight like a trans woman) Trava being a colloquial word from Argentina referring to trans woman. To fight is also fighting against prejudice and systematic violence.

“Amor trans”

(Trans love) - Expression of love.

“BEYAKO”

(Wild) - A Caribbean colloquial word for `Wicked` or `Lustful` and often used in Reggaeton music.

“MUAK”

An onomatopoeia of the sound of kissing.